

7. INTEGRAL METODLARI

u , x 'in türevlenebilir bir fonksiyonu ve $n \neq -1$ bir rasyonel sayı olmak üzere, zincir kuralından

$$\frac{d}{dx} \left(\frac{u^{n+1}}{n+1} \right) = u^n \frac{du}{dx} \quad \text{olduğunu biliyoruz.}$$

buradan

$$\int \left(u^n \frac{du}{dx} \right) dx = \frac{u^{n+1}}{n+1} + C \quad \text{elde edilir.}$$

Bu eşitliği aşağıdaki gibi ifade edebiliriz:

$$\int u^n du = \frac{u^{n+1}}{n+1} + C$$

Örnek. $\int \sqrt{1+y^2} \cdot 2y \, dy$ belirsiz integralini hesaplayınız.

Çözüm.

$$u = 1+y^2 \Rightarrow \frac{du}{dy} = 2y$$

$$\int \sqrt{1+y^2} \cdot 2y \, dy = \int \sqrt{u} \left(\frac{du}{dy} \right) dy = \int u^{1/2} du$$

$$= \frac{2}{3} u^{3/2} + C$$

$$= \frac{2}{3} (1+y^2)^{3/2} + C$$

Örnek. $\int \sqrt{4t-1} \, dt$ belirsiz integralini hesaplayınız.

Çözüm.

$$4t-1 = u \quad \Rightarrow \quad 4 \, dt = du$$

$$\int \sqrt{4t-1} \, dt = \int \sqrt{u} \, \frac{du}{4} = \frac{1}{4} \int u^{1/2} \, du$$

$$= \frac{1}{4} \frac{u^{3/2}}{3/2} + C = \frac{1}{6} u^{3/2} + C = \frac{1}{6} (4t-1)^{3/2} + C$$

Değişken Dönüşüm Kuralı

$u=g(x)$ değer kümesi bir I aralığı olan türevlenebilir bir fonksiyon ise ve f I üzerinde sürekli ise

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

olur.

örnek. $\int \sin^3 x \cos x \, dx = ?$

$$\sin x = u \Rightarrow \cos x \, dx = du$$

$$I = \int \sin^3 x \cos x \, dx = \int u^3 \, du = \frac{u^4}{4} + C = \frac{\sin^4 x}{4} + C$$

örnek. $\int \sin^3 x \, dx = ?$

$$\int \sin^2 x \sin x \, dx = \int (1 - \cos^2 x) \sin x \, dx = \underset{\substack{\downarrow \\ u = \cos x \, dx}}{-\int (1 - u^2) \, du}$$

$$= -u + \frac{u^3}{3} + C = -\cos x + \frac{\cos^3 x}{3} + C \quad -du = \sin x \, dx$$

drnek. $\int x \sqrt{2+3x} dx \underset{\downarrow}{=} \int \frac{u-2}{3} \cdot \sqrt{u} \frac{du}{3}$

$$u = 2+3x$$

$$du = 3 dx$$

$$x = \frac{u-2}{3}$$

$$= \frac{1}{9} \int \sqrt{u} (u-2) du = \frac{1}{9} \int (u^{3/2} - 2u^{1/2}) du$$

$$= \frac{1}{9} \cdot \frac{2}{5} u^{5/2} - \frac{2}{9} u^{3/2} \cdot \frac{2}{3} + C$$

$$= \frac{2}{45} (2+3x)^{5/2} - \frac{4}{27} (2+3x)^{3/2} + C$$

örnek. $\int \sec x \, dx = \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} \, dx$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} = \ln |\sec x + \tan x| + C$$

↓

$$u = \sec x + \tan x$$

$$du = (\tan^2 x + \sec x \tan x) \, dx$$

örnek. $\int \cos(7\theta + 5) d\theta = ?$

Çözüm. $u = 7\theta + 5, \quad du = 7 d\theta, \quad \frac{1}{7} du = d\theta$

$$\int \cos(7\theta + 5) d\theta = \int \cos u \frac{1}{7} du = \frac{1}{7} \int \cos u du$$

$$= \frac{1}{7} \sin u + C = \frac{1}{7} \sin(7\theta + 5) + C$$

Örnek. $\int x^2 \sin(x^3) dx = ?$

Çözüm. $u = x^3 \Rightarrow du = 3x^2 dx \Rightarrow x^2 dx = \frac{1}{3} du$

$$I = \int x^2 \sin(x^3) dx = \int \frac{1}{3} \sin u du = \frac{1}{3} \int \sin u du$$

$$= -\frac{1}{3} \cos u + C$$

$$= -\frac{1}{3} \cos(x^3) + C$$

örnek. $\int \frac{1}{\cos^2 2x} dx = ?$

Çözüm. $I = \int \sec^2 2x dx$, $2x = u$, $2dx = du$

$$I = \int \frac{1}{2} \sec^2 u du = \frac{1}{2} \int \sec^2 u du$$

$$= \frac{1}{2} \tan u + C$$

$$= \frac{1}{2} \tan(2x) + C$$

örnek. $\int \frac{2z \, dz}{\sqrt[3]{z^2+1}} = ?$

Çözüm. $z^2+1 = u$, $du = 2z \, dz$

$$I = \int \frac{du}{u^{1/3}} = \int u^{-1/3} \, du = \frac{u^{2/3}}{2/3} + C$$

$$= \frac{3}{2} u^{2/3} + C = \frac{3}{2} (z^2+1)^{2/3} + C$$

II. Çözüm yolu:

$$u = \sqrt[3]{z^2+1} \Rightarrow u^3 = z^2+1, \quad 3u^2 du = 2z dz$$

$$I = \int \frac{3u^2}{u} du = 3 \int u du = \frac{3u^2}{2} + C$$

$$= \frac{3}{2} (z^2 + 1)^{3/2} + C$$

örnek. $\int \sin^2 x \, dx = ?$, $\int \cos^2 x \, dx = ?$

Cözüm.

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx$$

$$= \frac{1}{2} \int (1 - \cos 2x) \, dx = \frac{1}{2} \int dx - \frac{1}{2} \int \cos 2x \, dx$$

$$= \frac{1}{2} x - \frac{1}{2} \frac{\sin 2x}{2} + C$$

$$= \frac{x}{2} - \frac{\sin 2x}{4} + C$$

TEOREM.

g' , $[a, b]$ aralığında sürekli ise ve f , g 'nin değer kümesinde sürekli ise

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du, \quad u = g(x)$$

olur.

örnek. $\int_{-1}^1 3x^2 \sqrt{x^3+1} \, dx = ?$

Çözüm.

$$x^3+1=u \quad \Rightarrow \quad 3x^2 \, dx = du$$

$$x=-1 \Rightarrow u=0$$

$$x=1 \Rightarrow u=2$$

$$I = \int_0^2 \sqrt{u} \, du = \frac{2}{3} u^{3/2} \Big|_0^2 = \frac{2}{3} (2^{3/2} - 0^{3/2})$$

$$= \frac{4\sqrt{2}}{3}$$

drnek. $\int_{\pi/4}^{\pi/2} \cot \theta \cdot \csc^2 \theta \, d\theta = ?$

$$u = \cot \theta, \quad du = -\csc^2 \theta \, d\theta, \quad \theta = \frac{\pi}{4} \Rightarrow u = 1$$

$$I = \int_1^0 u \cdot (-du)$$

$$\theta = \frac{\pi}{2} \Rightarrow u = 0$$

$$= \int_0^1 u \, du = \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}$$

örnek.

$$\int_1^e \frac{\ln x}{x} dx = \int_0^1 u du = \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}$$

$\downarrow$
 $u = \ln x$
 $\frac{1}{x} dx = du$

Veya

$$\int_1^e \frac{\ln x}{x} dx = \int_{x=1}^{x=e} u du = \frac{u^2}{2} \Big|_{x=1}^{x=e} = \left(\frac{(\ln x)^2}{2} \right) \Big|_{x=1}^{x=e}$$

$\downarrow$
 $u = \ln x$
 $\frac{1}{x} dx = du$

$$= \frac{1}{2} \left((\ln e)^2 - (\ln 1)^2 \right)$$
$$= \frac{1}{2}$$

Drnek.

$$\int \frac{dx}{\sqrt{x}(1+\sqrt{x})} = \int \frac{2du}{1+u}$$

$$\sqrt{x} = u$$

$$\frac{1}{2\sqrt{x}} dx = du$$

$$= 2 \ln |1+u| + C = 2 \ln |1+\sqrt{x}| + C$$

veya

$$\int \frac{dx}{\sqrt{x}(1+\sqrt{x})} = \int \frac{2u du}{u(1+u)} = 2 \ln |1+u| + C$$

$$= 2 \ln |1+\sqrt{x}| + C$$

$$x = u^2$$

$$dx = 2u du$$

$$* \int (2x+2) e^{x^2+2x+3} dx$$

$$x^2+2x+3 = u \Rightarrow (2x+2) dx = du$$

$$\int (2x+2) e^{x^2+2x+3} dx = \int e^u du = e^u + C$$

$$= e^{x^2+2x+3} + C$$

$$* \int (2x+5) (x^2+5x)^7 dx$$

$$x^2+5x=u \Rightarrow (2x+5)dx = du$$

$$I = \int u^7 du = \frac{u^8}{8} + C$$

$$= \frac{1}{8} (x^2+5x)^8 + C$$

$$* \int \frac{x^3}{(1+x^4)^{1/3}} dx = ?$$

$$u = 1 + x^4 \Rightarrow du = 4x^3 dx$$

$$\int \frac{x^3}{(1+x^4)^{1/3}} dx = \int \frac{1}{4} \frac{du}{u^{1/3}} = \frac{1}{4} \int u^{-1/3} du$$

$$= \frac{1}{4} u^{2/3} \cdot \frac{3}{2} + C$$

$$= \frac{3}{8} (1+x^4)^{2/3} + C$$

$$* \int \frac{\sin(\ln x)}{x} dx = ?$$

$$u = \ln x, \quad du = \frac{1}{x} dx$$

$$\begin{aligned} I &= \int \frac{\sin(\ln x)}{x} dx = \int \sin u \, du = -\cos u + C \\ &= -\cos(\ln x) + C \end{aligned}$$

$$* \int_0^9 \sqrt{4 - \sqrt{x}} \, dx = ?$$

$$u = 4 - \sqrt{x} \Rightarrow \sqrt{x} = 4 - u$$

$$x = (4 - u)^2 = u^2 - 8u + 16$$

$$dx = (2u - 8) \, du$$

$$x=0 \Rightarrow u=4$$

$$x=9 \Rightarrow u=1$$

$$\int_0^9 \sqrt{4-\sqrt{x}} \, dx = \int_4^1 \sqrt{u} \, (2u-8) \, du$$

$$= \int_4^1 u^{1/2} (2u-8) \, du = \int_4^1 (2u^{3/2} - 8u^{1/2}) \, du$$

$$= 2 \cdot \frac{2}{5} u^{5/2} - 8 \cdot \frac{2}{3} u^{3/2} \Big|_4^1$$

$$* \int \frac{(3 + \ln x)^2 (2 - \ln x)}{4x} dx = ?$$

$$u = 3 + \ln x \Rightarrow du = \frac{1}{x} dx, \quad \ln x = u - 3$$

$$I = \frac{1}{4} \int (3 + \ln x)^2 (2 - \ln x) \frac{1}{x} dx$$

$$= \frac{1}{4} \int u^2 (2 - (u - 3)) du$$

$$= \frac{1}{4} \int u^2 (2 - (u-3)) du$$

$$= \frac{1}{4} \int u^2 (5 - u) du$$

$$= \frac{1}{4} \left(5 \cdot \frac{u^3}{3} - \frac{u^4}{4} \right) + C$$

$$= \frac{5}{12} (3 + \ln x)^3 - \frac{1}{16} (3 + \ln x)^4 + C$$

$$* \int x \sqrt{4-x} \, dx$$

$$u = 4 - x \Rightarrow du = (-1) dx \quad , \quad -du = dx \quad , \quad x = 4 - u$$

$$I = \int (4 - u) \sqrt{u} \, (-1) \, du$$

$$= - \int (4 - u) \sqrt{u} \, du$$

$$= - \int (4 - u) u^{1/2} \, du = - \int (4 u^{1/2} - u^{3/2}) \, du$$

$$= - \int (4-u) u^{1/2} du = - \int (4 u^{1/2} - u^{3/2}) du$$

$$= - \left(4 \cdot \frac{2}{3} u^{3/2} - \frac{2}{5} u^{5/2} \right) + C$$

$$= - \left(\frac{8}{3} (4-x)^{3/2} - \frac{2}{5} (4-x)^{5/2} \right) + C$$

$$= \frac{2}{5} (4-x)^{5/2} - \frac{8}{3} (4-x)^{3/2} + C$$

$$* \int \frac{3}{x \ln x} dx = ?$$

$$u = \ln x \Rightarrow du = \frac{1}{x} dx, \quad I = 3 \int \frac{1}{u} du$$

$$= 3 \ln |u| + C$$

$$= 3 \ln |\ln x| + C$$

$$* \int_1^2 \left(\frac{2+\sqrt{x}}{x} \right)^{1/2} dx = ?$$

$$2 + \sqrt{x} = u \Rightarrow \frac{dx}{2\sqrt{x}} = du, \quad \frac{dx}{\sqrt{x}} = 2 du$$

$$\int_1^2 \frac{(2+\sqrt{x})^{1/2}}{\sqrt{x}} dx = \int_{x=1}^{x=2} u^{1/2} 2 du = \int_3^{2+\sqrt{2}} 2 u^{1/2} du$$

$$= 2 u^{3/2} \frac{2}{3} \Big|_3^{2+\sqrt{2}}$$

$$* \int e^{2x} \sqrt{e^x + 2} \, dx = ?$$

$$e^x + 2 = u \Rightarrow e^x \, dx = du$$

$$I = \int (u-2) \sqrt{u} \, du = \int (u^{3/2} - 2u^{1/2}) \, du$$

$$= \frac{2}{5} u^{5/2} - 2 \cdot \frac{2}{3} u^{3/2} + C$$

$$= \frac{2}{5} (e^x + 2)^{5/2} - \frac{4}{3} (e^x + 2)^{3/2} + C$$

NOT. f , $[-a, a]$ simetrik aralığı üzerinde sürekli olsun :

a. Eğer f çift ise
$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

b. Eğer tek ise
$$\int_{-a}^a f(x) dx = 0$$

olduğunu gösteriniz.

$$a. \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$= - \int_0^{-a} f(x) dx + \int_0^a f(x) dx$$

$$= - \int_0^a f(-u) (-du) + \int_0^a f(x) dx$$

$u = -x \Rightarrow du = -dx$

$x = 0 \Rightarrow u = 0$

$x = -a \Rightarrow u = a$

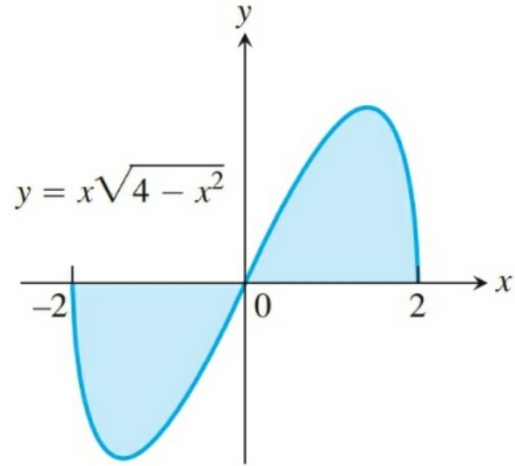
$$= \int_0^a f(u) du + \int_0^a f(x) dx = 2 \int_0^a f(x) dx$$

$\downarrow$
 $f(-u) = f(u)$

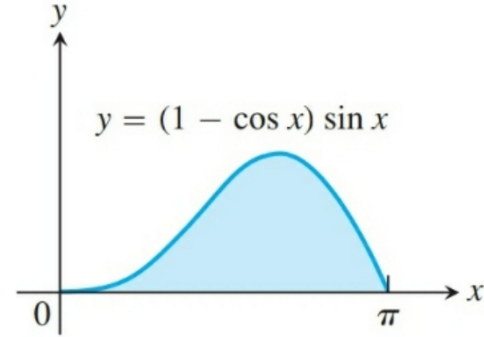
b. ODEV

Aşağıdaki alanları bulunuz.

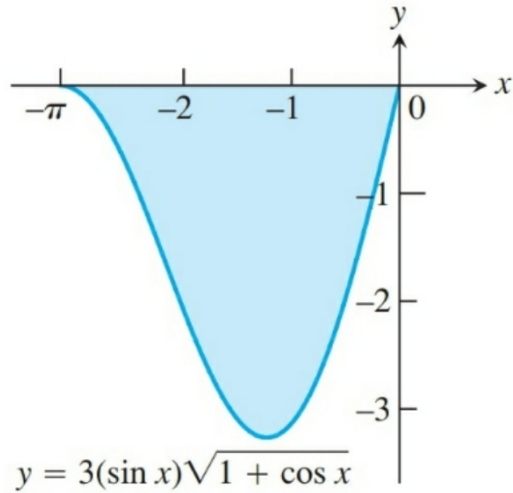
a.



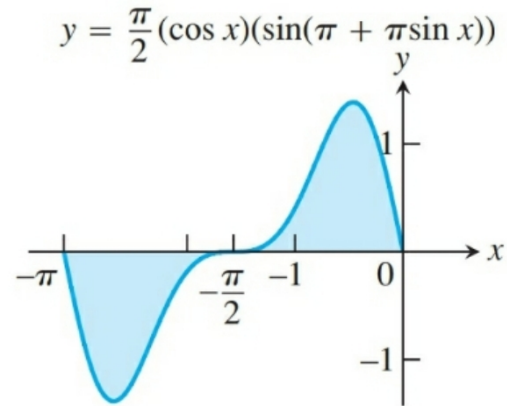
b.



c.



d.



Gözüm.

$$a. \quad A = \left| \int_{-2}^0 x \sqrt{4-x^2} \, dx \right| + \int_0^2 x \sqrt{4-x^2} \, dx$$

$$= - \underbrace{\int_{-2}^0 x \sqrt{4-x^2} \, dx}_{\substack{4-x^2=u \\ -2x \, dx = du}} + \underbrace{\int_0^2 x \sqrt{4-x^2} \, dx}_{\substack{4-x^2=u \\ -2x \, dx = du}}$$

$$= \frac{1}{2} \int_0^4 u^{1/2} \, du + \int_0^4 \frac{1}{2} u^{1/2} \, du = \int_0^4 u^{1/2} \, du = 16/3$$

$$b. \quad A = \int_0^{\pi} (1 - \cos x) \sin x \, dx$$

$$1 - \cos x = u \quad \Rightarrow \quad \sin x \, dx = du$$

$$x = 0 \quad \Rightarrow \quad u = 0$$

$$x = \pi \quad \Rightarrow \quad u = 2$$

$$A = \int_0^2 u \, du = \frac{u^2}{2} \Big|_0^2 = 2$$

c. $A = - \int_{-\pi}^0 3 \sin x \sqrt{1 + \cos x} \, dx$

$$1 + \cos x = u \quad \Rightarrow \quad -\sin x \, dx = du$$

$$x = 0 \quad \Rightarrow \quad u = 2$$

$$x = -\pi \quad \Rightarrow \quad u = 0$$

$$A = - \int_0^2 3 u^{1/2} (-du) = 3 \int_0^2 u^{1/2} du$$

$$= 2 u^{3/2} \Big|_0^2 = 2^{5/2}$$

d. Simetriden dolayı:

$$A = 2 \int_{-\frac{\pi}{2}}^0 \frac{\pi}{2} (\cos x) (\sin(\pi + \pi \sin x)) dx$$

$$= 2 \int_0^{\pi} \frac{\pi}{2} \sin u \frac{1}{\pi} du$$


$$u = \pi + \pi \sin x$$

$$du = \pi \cos x dx \quad \Rightarrow \quad \frac{1}{\pi} du = \cos x dx$$

$$x = -\frac{\pi}{2} \Rightarrow u = 0$$

$$x = 0 \Rightarrow u = \pi$$